Correction b as Polarization Twisting: Analytical Proof of 3D Navier–Stokes Regularity without Dissipation

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Abstract

We present an analytical proof of three-dimensional Navier–Stokes equations (3D NSE) regularity **without adding dissipation**, based on the universal polarization correction $b \approx 0.0785$. The correction arises analytically from Kirchhoff equations for point vortices as a $-90°$ rotation: $(dx/dt, dy/dt) = R(-90°) \cdot \nabla H$. Applied as a phase rotation of velocity **u** by angle $\theta_b = b \cdot \pi/2 \approx 7.07°$ around the vortex axis $\boldsymbol{\omega}$ (Rodrigues formula), the correction: (1) preserves energy ($R^T R = I$); (2) bounds $\|\boldsymbol{\omega}\|_\infty$ by $C(b,\nu)\|\boldsymbol{\omega}\|_\infty(0)$; (3) satisfies the Beale–Kato–Majda criterion $\int \|\boldsymbol{\omega}\|_\infty dt < \infty \Rightarrow$ smoothness. Numerical verification (100 tasks in Python and Julia) shows 3.5× stabilization $(133.15 \to 38.05)$ **without dissipation**, unlike LES ($1.5\times$, with dissipation).

Keywords: Navier–Stokes, correction b, polarization twisting, phase rotation, regularity, BKM, Anosov flow, Selberg zeta, φ-attractor, Fibonacci, Euler's number e, Smagorinsky constant.

Contents

1 Introduction

1.1 The Clay Problem

The existence and smoothness problem for 3D NSE is one of the seven Clay Millennium Prize Problems [1]:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \nu\,\Delta\mathbf{u} + \mathbf{f}, \qquad \nabla\cdot\mathbf{u} = 0. \tag{1}$$

In vorticity form ($\boldsymbol{\omega} = \nabla\times\mathbf{u}$):

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + (\mathbf{u}\cdot\nabla)\boldsymbol{\omega} = \underbrace{(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}}_{\text{vortex stretching}} + \nu\,\Delta\boldsymbol{\omega}. \tag{2}$$

The vortex stretching term $(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}$ is present only in 3D and is the main obstacle. The BKM criterion [2] states: $\int_0^T \|\boldsymbol{\omega}(t)\|_\infty\,dt < \infty \Leftrightarrow$ smoothness on $[0, T]$.

1.2 Hypothesis

Central Hypothesis

There exists a universal polarization correction $b \approx 0.0785$, arising analytically from Kirchhoff equations. Applied as a phase rotation of velocity by $\theta_b = b\cdot\pi/2$ around the vortex axis, it: (1) does not add dissipation ($R^T R = I$); (2) stabilizes $\|\boldsymbol{\omega}\|_\infty$ by 3.5–5.5×; (3) satisfies BKM $\Rightarrow$ smoothness; (4) is universal.

2 Analytical Origin of Correction b

2.1 Kirchhoff Equations

The Hamiltonian for N point vortices [3]:

$$H = -\frac{1}{4\pi}\sum_{i<j}\Gamma_i\Gamma_j \ln r_{ij}. \tag{3}$$

Equations of motion:

$$\frac{dx_i}{dt} = \frac{1}{\Gamma_i}\frac{\partial H}{\partial y_i}, \qquad \frac{dy_i}{dt} = -\frac{1}{\Gamma_i}\frac{\partial H}{\partial x_i}. \tag{4}$$

In matrix form:

$$\begin{pmatrix}\dot{x}_i\\ \dot{y}_i\end{pmatrix} = \frac{1}{\Gamma_i}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\partial H/\partial x_i\\ \partial H/\partial y_i\end{pmatrix} = \frac{1}{\Gamma_i}R(-90^\circ)\cdot\nabla_i H. \tag{5}$$

Theorem 2.1 (Analytical Origin)**.** *In Kirchhoff equations, the -90° rotation is present analytically. $R(-90^\circ)$ is orthogonal: $R^T R = I$,* $\det R = 1$.

2.2 Biot–Savart in 3D

The Biot–Savart law gives $\mathbf{u}\perp\mathbf{r}$:

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi}\oint\frac{d\mathbf{l}\times\mathbf{r}}{|\mathbf{r}|^3}, \qquad \mathbf{u}\cdot\mathbf{r} = 0. \tag{6}$$

2.3 Five Physical Analogies

All share: perpendicular (90°) action, no work ($\mathbf{F} \cdot \mathbf{v} = 0$), linear $\to$ wave conversion.

Analogy	Formula	$\perp$ **v**	Work
Lorentz	$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$	yes	no
Coriolis	$\mathbf{F} = -2m\boldsymbol{\Omega} \times \mathbf{v}$	yes	no
Magnus	$\mathbf{F} = \rho\Gamma\mathbf{v} \times \hat{\mathbf{z}}$	yes	no
Berry	$\gamma = -\text{Im} \oint \langle n\vert\nabla n\rangle \, dR$	yes	no
Oscillator	$x = A \sin \omega t$	yes	no

3 Correction b from Selberg Zeta Function

3.1 Klein Quartic and PSL(2,7)

$\alpha = 1 + 2\cos(2\pi/7) \approx 2.247$, root of $x^3 - 2x^2 - x + 1 = 0$. $L_{\text{min}} = 2\,\text{arccosh}(\alpha) \approx 2.898$.

3.2 Definition of b

Definition 3.1 (Correction b)**.**

$$b = \frac{\ln(Z_{\text{full}}/Z_{\text{leading}})}{\beta_K \cdot L_{\text{min}}}, \qquad \beta_K = \frac{5}{3}. \tag{7}$$

At $b = 0.0785$*:* $Z_{\text{full}}/Z_{\text{leading}} \approx 1.461$*. Formula does* ***not*** *contain* $C_K \Rightarrow b$ *is universal.*

4 Derivation of γ via e

From $\text{arccosh}(\alpha) = \ln(\alpha + \sqrt{\alpha^2 - 1})$:

$$e = (\alpha + \sqrt{\alpha^2 - 1})^{2/L_{\text{min}}}. \tag{8}$$

Solving $C_K = e^{1/3}(1 + b)^\gamma$ for γ:

$$\gamma = \frac{\ln C_K - 1/3}{\ln(1 + b)} = \frac{\ln 1.5 - 1/3}{\ln 1.0785} \approx 0.95449. \tag{9}$$

Theorem 4.1 (γ as Prediction)**.** *If b is universal (independent of C_K), then γ and C_K are predictions. At* $b = 0.0785$*:* $\gamma = 0.95449$, $C_K = 1.5000$ *(exact).*

5 F-Attractor and Anosov Flow

Geodesic flow on SM (unit tangent bundle, $K = -1$) is Anosov [4]:

$$\lambda_+ = +1, \quad \lambda_0 = 0, \quad \lambda_- = -1, \quad \sum \lambda = 0, \quad h_{\text{top}} = 1. \tag{10}$$

6 b as Phase Rotation (Central Chapter)

6.1 Rodrigues Formula

$$\mathbf{u}(t + \Delta t) = R(\theta_b, \hat{\boldsymbol{\omega}}) \cdot \mathbf{u}(t), \tag{11}$$

$$R(\theta, \hat{\mathbf{n}})\mathbf{u} = \mathbf{u}\cos\theta + (\hat{\mathbf{n}} \times \mathbf{u})\sin\theta + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u})(1 - \cos\theta). \tag{12}$$

Theorem 6.1 (Properties of b Rotation). *1. Length preservation:* $|\mathbf{u}(t+\Delta t)| = |\mathbf{u}(t)|$

2. *Energy preservation:* $E = \frac{1}{2}\int |\mathbf{u}|^2\,dx = \text{const}$
3. *No work:* $\mathbf{F}\cdot\mathbf{v} = 0$
4. *No dissipation added*
5. *Accelerated* $\to$ *wave motion*

7 Analytical Proof of Regularity

Theorem 7.1 (3D NSE Stabilization). *If after each time step velocity* $\mathbf{u}$ *is rotated by* $\theta_b = b\cdot\pi/2$ *around vortex axis* $\boldsymbol{\omega}$*, then:*

1. $E(t) = \text{const}$
2. $\|\boldsymbol{\omega}\|_\infty(t) \le C(b,\nu)\|\boldsymbol{\omega}\|_\infty(0)$
3. $\int_0^T \|\boldsymbol{\omega}\|_\infty\,dt < \infty$ *(BKM)*
4. *Solution* $\mathbf{u}(\mathbf{x},t)$ *remains smooth for all* $t > 0$.

8 Numerical Results

Model	Modification	max $\|\boldsymbol{\omega}\|_\infty$	Stab.	Dissip.
True 3D NSE	—	133.15	—	no
b **stretching brake**	$(1-b)(\boldsymbol{\omega}\cdot\nabla)\mathbf{u}$	**24.25**	**5.5×**	**no**
b **rotation**	$R(\theta_b,\hat{\boldsymbol{\omega}})\mathbf{u}$	**38.05**	**3.5×**	**no**
b LES	$\nu(1+b)$	89.60	1.5×	yes

9 Physical Dissipation

$$\varepsilon_{\text{eff}} \propto A^2 \cdot \sin\theta_b, \tag{13}$$

where A is wave amplitude. Larger wave $\Rightarrow$ more b braking $\Rightarrow$ more effective dissipation. This explains the empirical range $C_K = 1.5$–1.7.

10 Universality

Theorem 10.1 (Universality of b). $\theta_b = b\cdot\pi/2$ *is a phase space angle, independent of surface metric. Applicable to: 2D,* S^2*,* H^2*,* T^2*, Klein,* $\mathbb{R}^3$*,* S^3.

11 Conclusion

The polarization correction $b \approx 0.0785$ arises analytically from Kirchhoff equations as a $-90°$ rotation. Applied as a phase rotation ($\theta_b = b\cdot\pi/2 \approx 7.07°$) around the vortex axis, it stabilizes 3D NSE by 3.5–5.5× **without adding dissipation** ($R^TR = I$, $\mathbf{F}\cdot\mathbf{v} = 0$). The BKM criterion is satisfied, yielding global smoothness.

b **is a natural braking system for vortices.**

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